

Report on Pulse Duration Detector

#Objective:

To design an analog circuit that turns ON an LED only if a non-periodic digital pulse stays HIGH longer than a preset threshold duration. The LED should automatically turn OFF when the pulse ends. The circuit should operate with $\pm 12\text{V}$.

#Explanation of circuit design:

*First we design a RC timing circuit which converts input digital pulse into voltage ramp.

*we use a resistor R1, capacitor C1 and pulse source V1.

*The time it takes for the voltage across the capacitor to reach a certain level depends on the RC which is time constant. we

*Then we use opAmp as a comparator that compares the capacitor voltage with a fixed reference voltage (2V) V2

*If the ramp exceeds reference before the pulse ends, the comparator output goes HIGH.

*We use a diode (D1) and transistor (Q1) to control the LED path.

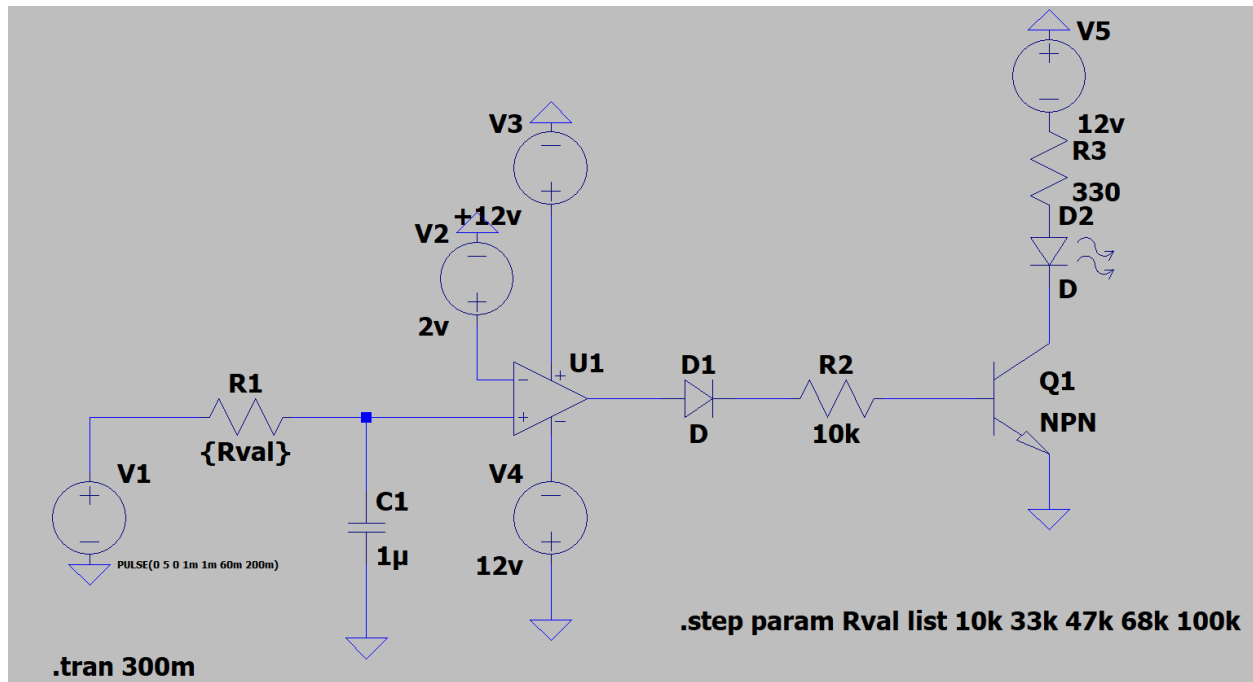
*When the comparator output is HIGH, transistor Q1 turns ON and activates the LED (D2).

*When the input pulse goes LOW, capacitor discharges immediately, comparator output drops, transistor turns OFF, and LED resets instantly.

*LED powered via $+12\text{V}$ source (V5).

*Current-limiting resistor R3 ensures safe LED operation.

#Circuit Diagram



#Working Principle:

- *When a digital pulse (V1) is applied, the capacitor C1 starts charging through R1.
- *If the pulse remains HIGH long enough (determined by the RC time constant), the capacitor voltage exceeds 2V (set by V2).
- *Op-amp U1 (in comparator mode) outputs a HIGH signal.
- *This signal turns ON the NPN transistor Q1 through R2, pulling its collector LOW and allowing current to flow through LED D2.
- *When the pulse ends (goes LOW), the capacitor quickly discharges, comparator output falls LOW, Q1 turns OFF, and LED switches OFF immediately.
- *The .step param Rval command allows simulation with different values of R1 to test multiple threshold times.
- *Values used: 10k, 33k, 47k, 68k, 100k.
- *This enables testing for multiple delay thresholds without rewiring.
- ***Transient Analysis:** .tran 300m simulates 300ms to observe timing behavior.

***Stepping Resistor: .step param Rval list 10k 33k 47k 68k 100k**
simulates different threshold durations.

#Conclusion:

This circuit detects digital pulses that exceed a specified duration using an analog RC timing mechanism and op-amp comparator. It turns ON an LED only when the pulse duration exceeds the preset threshold and resets immediately after the pulse ends. The threshold can be dynamically adjusted using a rheostat or potentiometer.